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3274 Comp. Methods

Homework 6

Chapter 5 Exercise 6

$$\varphi_n(x) = \frac{\pi^{-\frac{1}{4}}}{\sqrt{2^n n!}} e^{-\frac{x^2}{2}} H_n(x)$$

Are solutions to the harmonic oscillator problem with nondimensionalized Hamiltonian:

$$H = -\frac{1}{2} \frac{d^2}{dx^2} + \frac{1}{2} x^2$$

These aren't polynomials, but they are orthonormal functions under an inner product:

$$\langle f | g \rangle = \int_{-\infty}^{\infty} f(x) g(x) dx$$

- i) **Check** this orthonormality numerically by computing and printing out the matrix **A** where $(\mathbf{A})_{ij} \equiv \langle i | j \rangle = \langle \varphi_i(x) | \varphi_j(x) \rangle$, using Gauss-Hermite integration. Do this for $i, j = 0, 1 \dots 5$.
- ii) What is the minimum number of points needed for this computation?

Solution

We have a 6x6 matrix that's symmetric $\mathbf{A}_{ij} = \mathbf{A}_{ji}$, as the inner product is symmetric for real functions. I begin with initializing an Eigen Matrix (6x6). Then I fill in the matrix by integrating so:

$$(\mathbf{A})_{ij} = \int_{-\infty}^{\infty} \varphi_i(x) \varphi_j(x) dx$$

I choose to just do a 2D loop over the columns and rows.

To build each φ_n , I used QAT's generic functions. The normalized Hermite polynomial `HermitePolynomial(n, Genfun::TWIDDLE)` already includes the factor $\pi^{(-1/4)}/\sqrt{(2^n n!)}$, so we just multiply by the Gaussian to obtain:

$$\varphi_n(x) = e^{-\frac{x^2}{2}} \tilde{H}_n(x)$$

And we initialize new doubles in the matrices which are the returns of the integrator.

Findings

- i) With 6 Gauss-Hermite points, **A** became the 6x6 identity matrix. The off-diagonal entries are $\sim 10^{-16}$, which is just the floating point precision rounding error; thus they are orthogonal.
- ii) The minimum is 6 points.

$\varphi_i \varphi_j = \exp(-x^2) * (\text{degree } i+j \text{ polynomial})$. The $\exp(-x^2)$ is the Gauss-Hermite weight.

The N-point rule is exact up to degree $2N-1$. So with worst case $i = j = 5$ (degree 10),

$$2N-1 \geq 10 \Rightarrow N = 6$$

A run with $N = 5$ confirms it; every entry is right except $A_{55} \approx 0$. It should be 1.

```
(base) eliasdjipanyis@Eliass-MacBook-Pro-7 build % ./GAUSS-HERMITE
A (N = 6 points):
      1  1.21431e-16 -1.31839e-16 -4.51028e-17  4.96131e-16  6.245e-17
1.21431e-16      1  4.85723e-17  8.74301e-16  5.55112e-17 -1.66533e-16
-1.31839e-16  4.85723e-17      1  8.32667e-17  6.10623e-16  2.77556e-17
-4.51028e-17  8.74301e-16  8.32667e-17      1  5.55112e-17 -6.10623e-16
 4.96131e-16  5.55112e-17  6.10623e-16  5.55112e-17      1  1.11022e-16
 6.245e-17 -1.66533e-16  2.77556e-17 -6.10623e-16  1.11022e-16      1
```

Chapter 5 Exercise 7

The functions φ_i identified in the previous problem constitute a basis for a vector space of functions in the interval $[-\infty, \infty]$. Operators can be conveniently represented as matrices with inf dimensions. Using a basis we can compute finite dimensional submatrices whose elements

$$A_{ij} = \langle i | A | j \rangle = \langle i | A j \rangle$$

- a) Compute and print the lowest 5x5 submatrices starting at row 0 and column 0 for operator $\hat{x}$, in the basis of the previous problem.
- b) Repeat for the operator $\hat{x}^2$; also compare the matrix you obtain with the square of the matrix found in part (a) and explain any discrepancies.
- c) Repeat parts (a) and (b) for the operator $\hat{D}$.
- d) Compute and print the 5x5 submatrix, starting at row 0 and column 0, of the Hamiltonian operator. Do the elements make sense?

Solution

Since the φ_n form a basis, each operator is represented by a matrix with elements

$$(A)_{ij} = \langle \varphi_i | A \varphi_j \rangle = \int_{-\infty}^{\infty} \varphi_i(x) (A \varphi_j)(x) dx$$

I reuse the setup from Exercise 6 but apply the operator to φ_j before integrating against φ_i . In QAT: $\hat{x}\varphi_j = X^*\varphi_j$, $\hat{x}^2\varphi_j = X^*X^*\varphi_j$, $\hat{D}\varphi_j = \varphi_j.\text{prime}()$, and $\hat{D}^2\varphi_j = \varphi_j.\text{prime}().\text{prime}()$. I fill all the matrices in one double loop over $i, j = 0 \dots 4$. By linearity, $H_{ij} = -1/2 (D^2)_{ij} + 1/2 (x^2)_{ij}$. Entries below 10^{-12} are roundoff and are shown as 0.

Number of points: each integrand is e^{-x^2} times a polynomial. The worst case is $i = j = 4$ with a second-order operator, which gives degree 10. Since $2N-1 \geq 10$, $N = 6$.

(a) $\hat{x}$

```
0.0000 0.7071 0.0000 0.0000 0.0000
0.7071 0.0000 1.0000 0.0000 0.0000
0.0000 1.0000 0.0000 1.2247 0.0000
0.0000 0.0000 1.2247 0.0000 1.4142
0.0000 0.0000 0.0000 1.4142 0.0000
```

The matrix is symmetric and nonzero only next to the diagonal.

(b) $\hat{x}^2$ (left) vs $(\hat{x} \text{ matrix})^2$ (right)

```
0.5000 0.0000 0.7071 0.0000 0.0000  0.5000 0.0000 0.7071 0.0000 0.0000
0.0000 1.5000 0.0000 1.2247 0.0000  0.0000 1.5000 0.0000 1.2247 0.0000
0.7071 0.0000 2.5000 0.0000 1.7321  0.7071 0.0000 2.5000 0.0000 1.7321
0.0000 1.2247 0.0000 3.5000 0.0000  0.0000 1.2247 0.0000 3.5000 0.0000
0.0000 0.0000 1.7321 0.0000 4.5000  0.0000 0.0000 1.7321 0.0000 2.0000
```

The two match except the (4,4) entry, which differs by 2.5. The truncated 5×5 product stops at $k = 4$. The missing term is $x_{45} x_{54} = 5/2 = 2.5$.

(c) $\hat{D}$

```
0.0000 0.7071 0.0000 0.0000 0.0000
-0.7071 0.0000 1.0000 0.0000 0.0000
0.0000 -1.0000 0.0000 1.2247 0.0000
0.0000 0.0000 -1.2247 0.0000 1.4142
0.0000 0.0000 0.0000 -1.4142 0.0000
```

$\hat{D}$ has the same magnitudes as $\hat{x}$ but is antisymmetric. $\hat{D}^2$ and $(\hat{D} \text{ matrix})^2$ have diagonals of -0.5, -1.5, -2.5, -3.5, -4.5 and -0.5, -1.5, -2.5, -3.5, -2.0, so they differ at (4,4) by $D_{45} D_{54} = -2.5$, for the same truncation reason as in (b).

(d) $\hat{H}$

```
0.5000 0.0000 0.0000 0.0000 0.0000
0.0000 1.5000 0.0000 0.0000 0.0000
0.0000 0.0000 2.5000 0.0000 0.0000
0.0000 0.0000 0.0000 3.5000 0.0000
0.0000 0.0000 0.0000 0.0000 4.5000
```

This makes sense: the φ_n are eigenfunctions of $\hat{H}$, so the matrix is diagonal with entries $E_n = n + 1/2$. The off-diagonal terms of $-1/2 \hat{D}^2$ and $1/2 \hat{x}^2$ cancel exactly. Building $\hat{H}$ from the truncated squared matrices would instead give $H_{44} = 2.0$.

Chapter 5 Exercise 9

The charmonium system is a bound state of a charm quark and its antiquark, both treated as spinless, each with mass $m_c = 1370 \text{ MeV}/c^2$, in the potential

$$V(r) = -\alpha \frac{4\hbar c}{3r} + \frac{r}{\hbar c a^2}$$

with $\alpha = 0.38$ and $a = 2.43 \times 10^{-3} \text{ MeV}^{-1}$.

(a) Use the appropriate semiclassical quantization rule to find the energies of the S states ($\ell = 0$): the ground state and first two excited states.

(b) Do the same for the P-wave states ($\ell = 1$).

(c) Each state has a mass equal to the rest energy of the two quarks plus the excitation energy. Calculate this for the six states and plot an energy level diagram.

Solution

The two-body problem reduces to one particle with reduced mass $\mu = m_c/2 = 685 \text{ MeV}/c^2$. I choose to work in MeV and fm with $\hbar c = 197.327 \text{ MeV}\cdot\text{fm}$, so the Coulomb strength is $4\alpha\hbar c/3 \approx 100 \text{ MeV}\cdot\text{fm}$ and the string tension is $1/(\hbar c a^2) \approx 858 \text{ MeV}/\text{fm}$.

V_{eff} is a Genfun expression of a Variable R. Its minimum comes from a Bisection on $V_{\text{eff}}.\text{prime}()$, and the turning points from a Bisection on V_{eff} . The phase integral uses a GaussIntegrator with a 64-point GaussLegendreRule after substituting $r = r_1 + (r_2 - r_1)\sin^2\theta$, which avoids the square-root type behavior of p at the turning points. The energy is found with another Bisection on the quantization condition, wrapped in an F1D.

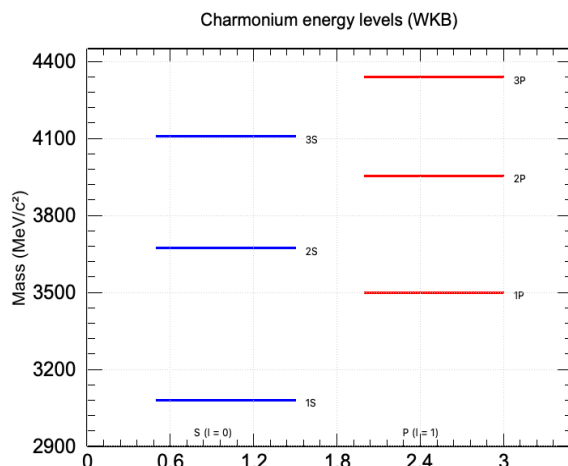
Results

state	E (MeV)	M = $2m_c + E$ (MeV)
1S	340.5	3080.5
2S	932.0	3672.0
3S	1367.1	4107.1
1P	758.8	3498.8
2P	1214.4	3954.4
3P	1598.7	4338.7

(a), (b): The S-state energies are 340.5, 932.0 and 1367.1 MeV, and the P-state energies are 758.8, 1214.4 and 1598.7 MeV.

(c): The excitation energies (340–1600 MeV) are 12–58% of the rest energy $2m_c c^2 = 2740 \text{ MeV}$, so they are significant.

Energy level diagram: S states (blue, left) and P states (red, right), plotted as mass $M = 2m_c c^2 + E$.



Chapter 6 Exercise 4

The Jones calculus applies to polarized light, but cannot handle partially polarized light. For these situations the polarization state is described by a four-dimensional Stokes vector, and optical elements by 4×4 Mueller matrices. Design and implement two classes implementing the Mueller calculus, together with a small test program, as was done for the Jones calculus.

Solution

The Stokes vector is built from intensities, so beams that are incoherent with each other simply add. That is what lets it describe partially polarized light:

$$S = (S_0, S_1, S_2, S_3) = (|E_x|^2 + |E_y|^2, |E_x|^2 - |E_y|^2, 2\text{Re}(E_x^* E_y), 2\text{Im}(E_x^* E_y))$$

$$P = \frac{\sqrt{S_1^2 + S_2^2 + S_3^2}}{S_0}, \quad S' = MS$$

P is the degree of polarization (0 for unpolarized, 1 for fully polarized light), and a chain of elements acts as the product of their Mueller matrices.

I designed the two classes, `StokesVector` and `MuellerMatrix`, to mirror `JonesVector` and `JonesMatrix`: header + .icc files in `MUELLERCALC/`, types (Horizontal, Vertical, Diagonal, Antidiagonal, Right, Left, plus Identity, FastHorizontal and FastVertical for the matrix), element access with `operator()`, `operator<<`, and `operator*` for matrix-matrix and matrix-vector products. Internally they store an `Eigen Vector4d` and `Matrix4d`, so the products use `Eigen`. New to the Mueller calculus are an Unpolarized Stokes vector, a Depolarizer matrix, `operator+` and scalar `*` for incoherent superposition, `intensity()` and `degreeOfPolarization()`, and `LinearPolarizer(theta)` for a polarizer at any angle.

Results

- 1) Circular polarizer (`FastHorizontal*Diagonal*FastVertical`), horizontal light in
Initial= [1,1,0,0] intensity= 1 DOP= 1
Final= [0.5,0,0,0.5] intensity= 0.5 DOP= 1
- 2) Unpolarized light through a horizontal polarizer
Initial= [1,0,0,0] intensity= 1 DOP= 0
Final= [0.5,0.5,0,0] intensity= 0.5 DOP= 1
- 3) Malus's law: I = 0.375, 0.25, 0.125, 0 at theta = 30, 45, 60, 90 deg
- 4) 60% unpolarized + 40% horizontal: [1,0.4,0,0], DOP= 0.4
Horizontal polarizer: I= 0.7 Vertical polarizer: I= 0.3
- 5) Unpolarized through H,V: I= 0 through H,diagonal,V: I= 0.125
- 6) Horizontal light through a depolarizer: [1,0,0,0], I= 1, DOP= 0

Test 1 shows horizontal light through the circular polarizer comes out right circular ($S_3 = S_0$) with half the intensity. Tests 2–6 need the Mueller calculus. Unpolarized light gets polarized by a polarizer and loses half its intensity, Malus's law $I = \frac{1}{2}\cos^2\theta$ is reproduced exactly, and the partially polarized beam has $P = 0.4$ and splits 0.7 / 0.3 between horizontal and vertical. Inserting a diagonal polarizer between crossed polarizers transmits 1/8 of the light, and the depolarizer keeps the intensity while setting $P = 0$. None of tests 2, 4 and 6 can be written as a Jones vector.